

Solid Waste Management and

□ To enable students to familiarize with various functional elements involved in Solid

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6351B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6351B.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Civil and EnvironmentalEngineering
Course code	6351B
Course title	Solid Waste Management and
Semester	6 Credits: 4
Course category	Programme Elective Course
Periods per week	4(L:4 T:0 P:0) Periods per semester:60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To enable students to familiarize with various functional elements involved in Solid Waste management.
- □ To enable students to discuss various methods for solid waste disposal.
- □ To enable students to identify impacts of air pollution and discuss various methods for air pollution control.

Course outcomes

CO	Official outcome	Study level
CO1	and characteristics and identify steps in waste 14 collection, storage and transportation. Discuss various methods for disposal of solid waste Understanding	See official syllabus
CO2	15 based on waste characteristics. Identify sources, impacts and sampling and Understanding	See official syllabus
CO3	15 measurement methods of air pollutants.	See official syllabus
CO4	Discuss various methods for air pollution control. 14 Understanding Tests 2	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 2 3
CO2	CO2 3 2 2 3
CO3	CO3 3 2 2 3

Row	Extracted official mapping
CO4	CO4 3 2 2 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	in waste collection, storage and transportation.	4	As specified
Module 2	characteristics.	3	As specified
Module 3	pollutants.	4	As specified
Module 4	Discuss various methods for air pollution control.	4	As specified

MODULE 1

in waste collection, storage and transportation.

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: in waste collection, storage and transportation.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Classify solid waste based on source. 1 Understanding Describe characteristics of each category of

Classify solid waste based on source. 1 Understanding Describe characteristics of each category of is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify solid waste based on source. 1 Understanding Describe characteristics of each category of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify solid waste based on source. 1 understanding describe characteristics of each category of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify solid waste based on source. 1 Understanding Describe characteristics of each category of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രകൃതി സംരക്ഷണം Classify solid waste based on source. 1 Understanding Describe characteristics of each category of പര്യവേക്ഷണ പ്രകാരം definition/meaning പ്രകാരം. പര്യവേക്ഷണ പ്രകാരം, steps, application പ്രകാരം പ്രകാരം പ്രകാരം. Technical terms English-പ്രകാരം പ്രകാരം പ്രകാരം.

5 Understanding waste. Identify purpose and functional elements of solid

5 Understanding waste. Identify purpose and functional elements of solid is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding waste. Identify purpose and functional elements of solid using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding waste. identify purpose and functional elements of solid should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding waste. Identify purpose and functional elements of solid means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-5 Understanding waste. Identify purpose and functional elements of solid waste management definition/meaning of waste management. Waste management steps, application of waste management. Technical terms English- waste management.

3 Understanding waste management. Describe waste collection, storage and

3 Understanding waste management. Describe waste collection, storage and is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding waste management. Describe waste collection, storage and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding waste management. describe waste collection, storage and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding waste management. Describe waste collection, storage and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-3 Understanding waste management. Describe waste collection, storage and പ്രവർത്തനം definition/meaning പ്രവർത്തനം . പ്രവർത്തനം പ്രവർത്തനം പ്രവർത്തനം , steps, application പ്രവർത്തനം പ്രവർത്തനം പ്രവർത്തനം . Technical terms English- പ്രവർത്തനം പ്രവർത്തനം .

5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste

5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes.

Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding transportation system total 14 introduction to the course wastes-classification based on source- municipal,industrial, bio-medical, universal, construction and demolition,radioactive, e wastes, agricultural waste- characteristics of each type, necessity of safe disposal of solid wastes. hazardous waste- identification and classification of hazardous waste. solid waste management- purpose, functional elements, hierarchy of waste management options. solid waste management rules 2016 waste collection, storage and transport: collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. processing techniques- mechanical volume and size reduction,chemical volume reduction, component separation. discuss various methods for disposal of solid waste based on waste should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding transportation system Total 14 Introduction to the course Wastes- Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-5 Understanding transportation system Total 14
 Introduction to the course Wastes-Classification based on source-
 Municipal,Industrial, Bio-medical, Universal, Construction and
 demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type,
 necessity of safe disposal of solid wastes. Hazardous waste- Identification and
 classification of hazardous waste. Solid waste management- Purpose, Functional
 elements, hierarchy of waste management options. Solid Waste management rules
 2016 Waste Collection, Storage and Transport: Collection
 components,storagecontainers/collection vehicles, collection operation, transfer
 station, waste collection system design, record keeping, control, inventory and
 monitoring, implementing collection and transfer system. Processing techniques-
 Mechanical volume and size reduction,chemical volume reduction, component
 separation. Discuss various methods for disposal of solid waste based on waste
 definition/meaning .
 steps, application . Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

in waste collection, storage and transportation.

M1.01	Classify solid waste based on source.	1
Understanding		
	Describe characteristics of each category of	
M1.02		5
Understanding	waste.	
	Identify purpose and functional elements of solid	
M1.03		3
Understanding	waste management.	
	Describe waste collection, storage and	
M1.04		5
Understanding	transportation system	
	Total	14

Contents:

Introduction to the course

Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal,

Construction and demolition,Radioactive, e wastes, Agricultural waste-

Characteristics of

each type, necessity of safe disposal of solid wastes.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

characteristics.

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: characteristics.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

6 Understanding solid wastes. Discuss thermo chemical methods for disposal of

6 Understanding solid wastes. Discuss thermo chemical methods for disposal of is an official topic in Module 2 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding solid wastes. discuss thermo chemical methods for disposal of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാഠ്യ 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of പൊതുവെ ഉപയോഗിക്കാൻ പറ്റാത്ത വസ്തുക്കൾ definition/meaning പൊതുവെ ഉപയോഗിക്കാൻ പറ്റാത്ത വസ്തുക്കൾ. പൊതുവെ ഉപയോഗിക്കാൻ പറ്റാത്ത വസ്തുക്കൾ, steps, application പൊതുവെ ഉപയോഗിക്കാൻ പറ്റാത്ത വസ്തുക്കൾ. Technical terms English-പൊതുവെ ഉപയോഗിക്കാൻ പറ്റാത്ത വസ്തുക്കൾ.

6 Understanding solid wastes.

6 Understanding solid wastes. is an official topic in Module 2 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding solid wastes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding solid wastes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding solid wastes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-6 Understanding solid wastes. **മുഖ്യപ്പെടുത്തേണ്ട**
മുഖ്യപ്പെടുത്തേണ്ട definition/meaning **മുഖ്യപ്പെടുത്തേണ്ട**. **മുഖ്യപ്പെടുത്തേണ്ട** **മുഖ്യപ്പെടുത്തേണ്ട**, steps,
application **മുഖ്യപ്പെടുത്തേണ്ട** **മുഖ്യപ്പെടുത്തേണ്ട** **മുഖ്യപ്പെടുത്തേണ്ട**. **Technical terms English-മുഖ്യപ്പെടുത്തേണ്ട**
മുഖ്യപ്പെടുത്തേണ്ട.

**Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1
Total 16 Contents Sanitary landfills- parts and their functions, methods of
landfilling, advantages and disadvantages. Composting- Stages in aerobic
composting, types of composting-Indore and Bangalore process Anaerobic
digesters – Stages in anaerobic digestion, Parts of a digester, Products of
anaerobic digestion, sludge disposal methods Thermo chemical methods -
incineration, gasification and pyrolysis. Types of incinerators - parts of an
incinerator, effluent gas and composition, advantages and disadvantages.
Hazardous waste management-Handling, Storage, Collection, and
Transportation of hazardous waste. Treatment and remedial actions-
Stabilization and Solidification, Thermal methods, Secure Landfill. Identify
sources, impacts and sampling and measurement methods of air**

Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1
Total 16 Contents Sanitary landfills- parts and their functions, methods of
landfilling, advantages and disadvantages. Composting- Stages in aerobic
composting, types of composting-Indore and Bangalore process Anaerobic
digesters – Stages in anaerobic digestion, Parts of a digester, Products of
anaerobic digestion, sludge disposal methods Thermo chemical methods -
incineration, gasification and pyrolysis. Types of incinerators - parts of an
incinerator, effluent gas and composition, advantages and disadvantages.
Hazardous waste management-Handling, Storage, Collection, and Transportation
of hazardous waste. Treatment and remedial actions- Stabilization and
Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and
sampling and measurement methods of air is an official topic in Module 2 of Solid
Waste Management and. Study the terminology first, then connect the stated
purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss treatment methods for hazardous wastes 3
Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their
functions, methods of landfilling, advantages and disadvantages. Composting-
Stages in aerobic composting, types of composting-Indore and Bangalore process
Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of
anaerobic digestion, sludge disposal methods Thermo chemical methods -

incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages.

Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss treatment methods for hazardous wastes 3 understanding series 1 1 total 16 contents sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. composting- stages in aerobic composting, types of composting-indoreand bangalore process anaerobic digesters - stages in anaerobic digestion, parts of a digester, products of anaerobic digestion, sludge disposal methods thermo chemical methods -incineration, gasification and pyrolysis. types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. hazardous waste management-handling, storage, collection, and transportation of hazardous waste. treatment and remedial actions- stabilization and solidification, thermal methods, secure landfill. identify sources, impacts and sampling and measurement methods of air should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indoreand Bangalore process Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indore and Bangalore process Anaerobic digesters - Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods - incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air pollution. definition/meaning. steps, application. Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

characteristics.

	Discuss bio chemical methods for disposal of	
M2.01		6
Understanding	solid wastes.	
	Discuss thermo chemical methods for disposal of	
M2.02		6
Understanding	solid wastes.	
M2.03	Discuss treatment methods for hazardous wastes	3
Understanding		
	Series 1	1
	Total	16

Contents

Sanitary landfills- parts and their functions, methods of landfilling, advantages and

disadvantages.

Composting- Stages in aerobic composting, types of composting-Indore and Bangalore process

Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of

anaerobic digestion, sludge disposal methods

Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators -

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

pollutants.

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: pollutants.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Understanding pollutants. Identify effects of air pollutants of environment

6 Understanding pollutants. Identify effects of air pollutants of environment is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding pollutants. Identify effects of air pollutants of environment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding pollutants. identify effects of air pollutants of environment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding pollutants. Identify effects of air pollutants of environment means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 6 Understanding pollutants. Identify effects of air pollutants of environment പ്ലഗ്ഗട്ട് ഡ്ലഗ്ഗട്ട് definition/meaning പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട്. പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട്, steps, application പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട്. Technical terms English- പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട് പ്ലഗ്ഗട്ട്.

4 Understanding and health. Discuss sources and impacts of indoor air

4 Understanding and health. Discuss sources and impacts of indoor air is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and health. Discuss sources and impacts of indoor air using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and health. discuss sources and impacts of indoor air should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and health. Discuss sources and impacts of indoor air means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-4 Understanding and health. Discuss sources and impacts of indoor air pollution. definition/meaning of indoor air pollution. sources, steps, application of indoor air pollution. Technical terms English- Malayalam.

2 Understanding pollution. Explain air sampling and pollution measurement

2 Understanding pollution. Explain air sampling and pollution measurement is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding pollution. Explain air sampling and pollution measurement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding pollution. explain air sampling and pollution measurement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding pollution. Explain air sampling and pollution measurement means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 2 Understanding pollution. Explain air sampling and pollution measurement ഏർപ്പെടുത്തൽ നിർവ്വചനം definition/meaning ഏർപ്പെടുത്തൽ. ഏർപ്പെടുത്തൽ ഏർപ്പെടുത്തൽ, steps, application ഏർപ്പെടുത്തൽ ഏർപ്പെടുത്തൽ. Technical terms English- ഏർപ്പെടുത്തൽ ഏർപ്പെടുത്തൽ.

3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.

3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding methods. total 15 modern air pollutants- classification of pollutants, important gaseous pollutants and sources of pollutants. bio pollutants. effects of air pollutants on atmosphere- reactions of pollutants in the atmosphere and their effects- smoke, smog and ozone layer disturbance and greenhouse effect, acid rain. effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. indoor air pollution- sources, types and health impacts. air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air

sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. പദപരിഭാഷ definition/meaning പദപരിഭാഷ. പദപരിഭാഷ പദപരിഭാഷ, steps, application പദപരിഭാഷ പദപരിഭാഷ. Technical terms English-പദപരിഭാഷ പദപരിഭാഷ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

pollutants.

	Identify sources, characteristics of modern air	
M3.01		6
Understanding	pollutants.	
	Identify effects of air pollutants of environment	
M3.02		4
Understanding	and health.	
	Discuss sources and impacts of indoor air	
M3.03		2
Understanding	pollution.	
	Explain air sampling and pollution measurement	
M3.04		3
Understanding	methods.	
	Total	15

Contents:

Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and

sources of pollutants. Bio pollutants.

Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their

effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Discuss various methods for air pollution control.

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Discuss various methods for air pollution control.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding for air quality monitoring. Discuss various control methods for particulate

2 Understanding for air quality monitoring. Discuss various control methods for particulate is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding for air quality monitoring. Discuss various control methods for particulate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding for air quality monitoring. discuss various control methods for particulate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding for air quality monitoring. Discuss various control methods for particulate means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 2 Understanding for air quality monitoring.

Discuss various control methods for particulate മലിനീകരണ പദാർത്ഥങ്ങൾ
definition/meaning പദാർത്ഥങ്ങൾ. പദാർത്ഥങ്ങൾ പദാർത്ഥങ്ങൾ, steps, application
പദാർത്ഥങ്ങൾ പദാർത്ഥങ്ങൾ പദാർത്ഥങ്ങൾ. Technical terms English-ൽ പദാർത്ഥങ്ങൾ പദാർത്ഥങ്ങൾ.

5 Understanding pollutants. Discuss various control methods for gaseous

5 Understanding pollutants. Discuss various control methods for gaseous is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding pollutants. Discuss various control methods for gaseous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding pollutants. discuss various control methods for gaseous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding pollutants. Discuss various control methods for gaseous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-5 Understanding pollutants. Discuss various control methods for gaseous pollutants. definition/meaning of pollutants. steps, application of control methods. Technical terms English- Malayalam.

5 Understanding pollutants. Discuss various biological air pollution control

5 Understanding pollutants. Discuss various biological air pollution control is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding pollutants. Discuss various biological air pollution control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding pollutants. discuss various biological air pollution control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding pollutants. Discuss various biological air pollution control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-5 Understanding pollutants. Discuss various biological air pollution control **ജൈവ വായു മലിനീകരണ നിയന്ത്രണ** definition/meaning **അർത്ഥം/അർത്ഥം**. **ജൈവ വായു മലിനീകരണ നിയന്ത്രണ**, steps, application **പ്രയോഗം** **ജൈവ വായു മലിനീകരണ നിയന്ത്രണ** **ജൈവ വായു മലിനീകരണ നിയന്ത്രണ**. Technical terms English-**ജൈവ വായു മലിനീകരണ നിയന്ത്രണ**.

2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/>

2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. - Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> is an official topic in Module 4 of Solid

Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., "Environmental Engineering Vol. II", Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, "Environmental Pollution Control Engineering" , New Age T2 International Publishers, 2006 T3 R C Gaur, "Basic Environmental Engineering", New Age Publications, 2008 M. N. Rao, H. V. N. Rao, "Air pollution", Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, " Environmental Engineering" , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, " Air Pollution Control Engineering", Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding techniques. series 2 1 total 15 ambient air quality and emission standards in india, air pollution indices, air act legislations and regulations. removal of gaseous pollutants by adsorption, absorption, condensation and combustion- principle, working, advantages and disadvantages of each method. particulate emission control by settling chambers, cyclone separation, wet collectors, fabric filters, electrostatic precipitators. principle, working, advantages and disadvantages of each method. biological air pollution control technologies. -biofilters,

bio trickling filters and bio scrubbers. text / reference: t/r book title/author garg, s.k., “environmental engineering vol. ii”, khanna publishers, new delhi t1 , latest edition c. s rao, “environmental pollution control engineering” , new age t2 international publishers, 2006 t3 r c gaur, “basic environmental engineering”, new age publications, 2008 m. n. rao, h. v. n. rao, “air pollution”, tata mcgraw hill pvt. ltd, new delhi r1 2017 cpheeo. manual on municipal solid waste management. central public health r2 and environmental engineering organization, government of india, new delhi., 2016 peavy s.w., rowe d.r. and tchobanoglous g, “ environmental engineering” , r3 mcgraw hill,new delhi, 1985. keshav kant, “ air pollution control engineering”, khanna publishing house, r4 2023 online resources: sl.no. website link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill,New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 https://cpcb.nic.in/ 2 https://nptel.ac.in/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers,

cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., "Environmental Engineering Vol. II", Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, "Environmental Pollution Control Engineering" , New Age T2 International Publishers, 2006 T3 R C Gaur, "Basic Environmental Engineering", New Age Publications, 2008 M. N. Rao, H. V. N. Rao, "Air pollution", Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, " Environmental Engineering" , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, " Air Pollution Control Engineering", Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Discuss various methods for air pollution control.

	Identify air pollution indices available in India	
M4.01		2
Understanding	for air quality monitoring.	
	Discuss various control methods for particulate	
M4.02		5
Understanding	pollutants.	
	Discuss various control methods for gaseous	
M4.03		5
Understanding	pollutants.	
	Discuss various biological air pollution control	
M4.04		2
Understanding	techniques.	
	Series 2	1
	Total	15

Contents:

Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations.
Removal of gaseous pollutants by adsorption, absorption, condensation and

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Classify solid waste based on source. 1 Understanding Describe characteristics of each category of. (3 marks)

Answer: Begin with the standard definition or identity of *Classify solid waste based on source. 1 Understanding Describe characteristics of each category of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding waste. Identify purpose and functional elements of solid. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding waste*. *Identify purpose and functional elements of solid*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding waste management. Describe waste collection, storage and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding waste management*. *Describe waste collection, storage and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical

volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste. (3 marks)

Answer: Begin with the standard definition or identity of 5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of. (3 marks)

Answer: Begin with the standard definition or identity of 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 6 Understanding solid wastes.. (3 marks)

Answer: Begin with the standard definition or identity of 6 Understanding solid wastes.. State the governing principle, list the key components or stages, and close with one

relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indore and Bangalore process Anaerobic digesters - Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indore and Bangalore process Anaerobic digesters - Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management- Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 6 Understanding pollutants. Identify effects of air pollutants of environment. (3 marks)

Answer: Begin with the standard definition or identity of 6 *Understanding pollutants*. Identify effects of air pollutants of environment. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding and health. Discuss sources and impacts of indoor air. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding and health*. Discuss sources and impacts of indoor air. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding pollution. Explain air sampling and pollution measurement. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding pollution*. Explain air sampling and pollution measurement. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone

layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding methods*. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 2 Understanding for air quality monitoring. Discuss various control methods for particulate. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding for air quality monitoring*. Discuss various control methods for particulate. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding pollutants. Discuss various control methods for gaseous. (3 marks)

Answer: Begin with the standard definition or identity of 5 *Understanding pollutants*. Discuss various control methods for gaseous. State the governing principle, list the key

components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding pollutants. Discuss various biological air pollution control. (3 marks)

Answer: Begin with the standard definition or identity of 5 *Understanding pollutants*. *Discuss various biological air pollution control*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., "Environmental Engineering Vol. II", Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, "Environmental Pollution Control Engineering" , New Age T2 International Publishers, 2006 T3 R C Gaur, "Basic Environmental Engineering", New Age Publications, 2008 M. N. Rao, H. V. N. Rao, "Air pollution", Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, " Environmental Engineering" , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, " Air Pollution Control Engineering", Khanna Publishing House, R4 2023 Online resources: SI.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/>. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding techniques*. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Classify solid waste based on source. 1 Understanding Describe characteristics of each category of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **6 Understanding solid wastes. Discuss thermo chemical methods for disposal of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **6 Understanding pollutants. Identify effects of air pollutants of environment.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Understanding for air quality monitoring. Discuss various control methods for particulate.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No. Website Link <https://cpcb.nic.in/> <https://nptel.ac.in/>

IN-PAGE SEARCH

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